@codedsoul_05

COMPUTER NETWORKS

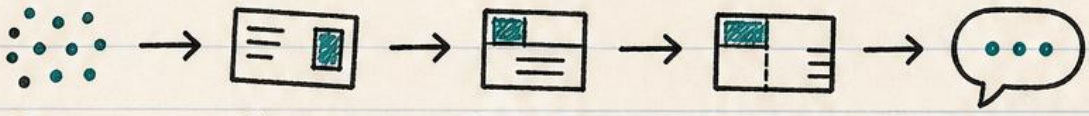
OSI → TCP/IP → PROTOCOLS

— COMPLETE PREMIUM NOTES —

Layers • IP • TCP/UDP • DNS • HTTP

Classic study notes — how machines talk across the wire.

A network moves bits with agreements — layers, addresses, and reliable delivery when you need it.



KEY TOPICS: OSI • IP • SUBNET • ROUTING • TCP • DNS • HTTP • SECURITY

WHAT'S INSIDE

- ✓ OSI / TCP-IP model
- ✓ Ethernet & framing
- ✓ IP & subnetting
- ✓ Routing (static & dynamic)
- ✓ TCP / UDP
- ✓ DNS
- ✓ HTTP / TLS
- ✓ NAT / Firewall
- ✓ Interview questions

BY THE NUMBERS

20+ pages of crisp notes

Diagrams that make it click

Exam patterns covered

Common mistakes to avoid

TOPICS

- ✓ Layers & encapsulation
- ✓ Addressing (MAC / IP)
- ✓ Transport (TCP / UDP)
- ✓ Application (DNS / HTTP)

Think in layers. Address with intent.

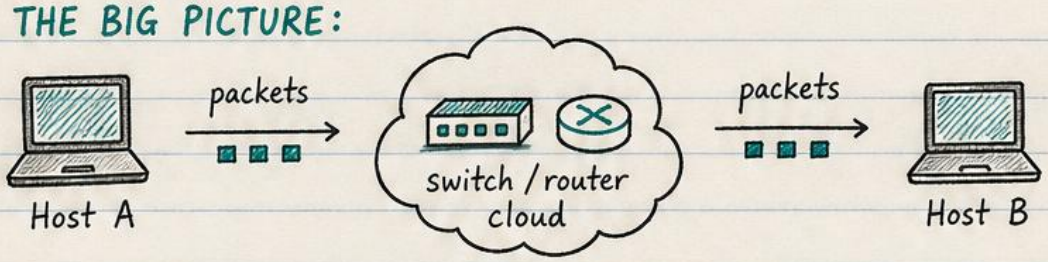
01 — WHAT & WHY NETWORKS

① **DEFINITION:** A computer network interconnects devices so they can exchange data — LANs, WANs, the Internet.

② **WHY study:**

- every app is networked.
- interviews love OSI/TCP.
- debugging "it works on my machine" often means the network path.

③ **THE BIG PICTURE:**



④ **GOALS of NETWORKING:**

connectivity | reliability | performance | security | scalability

⑤ **TYPES:**

- By Scope: LAN (Local Area Network)
  MAN (Metropolitan Area Network)
  WAN (Wide Area Network)
- By Architecture: Client-Server → server provides services, clients request.
  P2P (Peer-to-Peer) → peers share resources directly, no central server.

⑥ **INTERVIEW Q:** What is a protocol?

→ An agreed set of rules that define how data is formatted, ordered, transmitted, and how errors are handled between communicating entities.

⑦ **COMMON MISTAKE:** Memorizing layer numbers (7 layers? 5 layers?)

→ without knowing what each layer actually does. Understand the function, not just the position.

✶ **TAKEAWAY:** Networks are shared agreements + paths for data.

02 — OSI MODEL (7 LAYERS)

① OSI = conceptual 7-layer model for how network communication is organized.

②

7	Application	— user apps & APIs
6	Presentation	— format, encryption, compression
5	Session	— dialogues / sessions
4	Transport	— end-to-end reliability (TCP/UDP)
3	Network	— routing & IP addressing
2	Data Link	— frames, MAC, local delivery
1	Physical	— bits on wire/radio

TOP (closest to user)

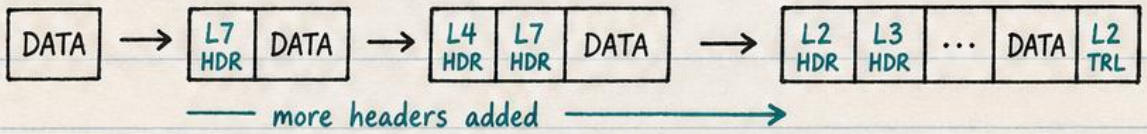
BOTTOM (closest to medium)

③ Mnemonic tip (common exam mnemonic):

"Please Do Not Throw Sausage Pizza Away"

(bottom-up)

④ Encapsulation idea: as data goes down the stack, each layer adds its header (and sometimes trailer).



⑤ Interview point: OSI is a teaching model; real Internet uses TCP/IP (fewer layers).

⑥ Common mistake: saying "HTTP is layer 3" — HTTP is an Application layer protocol (Layer 7).

TAKEAWAY: Seven jobs. One stack. Know each job's question.

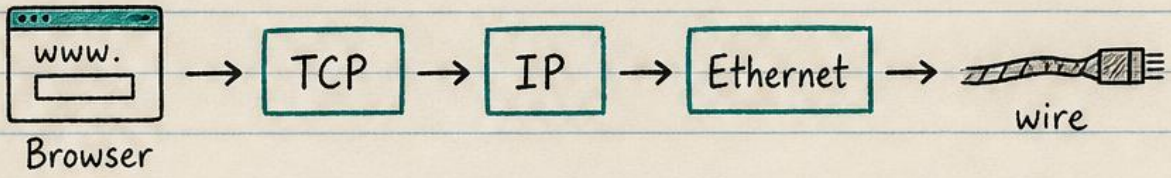
03 — TCP/IP MODEL

1. Practical Internet model — usually 4 (sometimes 5) layers.

2. Mapping table OSI → TCP/IP:

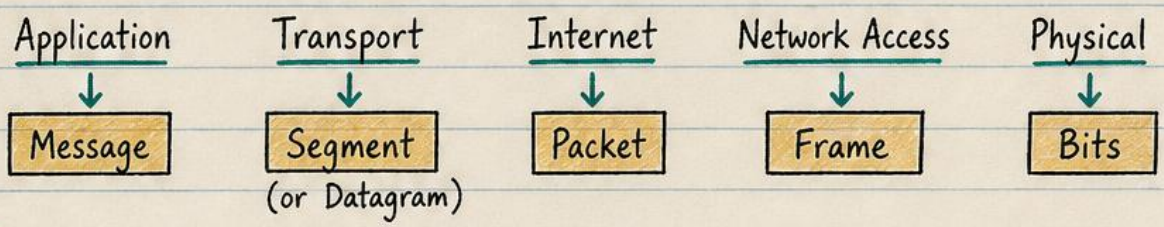
OSI (7-layer)	TCP/IP Model	Examples / Protocols
App + Pres + Sess	→ Application	(HTTP, DNS, SMTP...)
Transport	→ Transport	(TCP/UDP)
Network	→ Internet	(IP, ICMP, routing)
Data Link + Physical	Network Access / Link	(Ethernet, Wi-Fi, etc.)

3. Tiny sketch (data flow):



4. Why it matters: this is what you actually configure and debug.

5. PDU names (as data moves down the stack):



6. Interview point:
   - Compare OSI vs TCP/IP — OSI is theoretical; TCP/IP is implemented.

7. Common mistake:
   - Treating layer names as interchangeable without mapping.

★ TAKEAWAY: "OSI teaches. TCP/IP runs the Internet."

04 — PHYSICAL & DATA LINK

① PHYSICAL (L1)

bits as signals

- cables (copper / fiber)



- Wi-Fi radio (waves)



- hubs (legacy)



- bandwidth vs latency
  - bandwidth: how much data per second
  - latency: how long until it starts / arrives

② DATA LINK (L2)

- frames
- MAC addresses (48-bit hardware IDs)
- Ethernet (IEEE 802.3)
- Switches forward by MAC (learn & forward)
- collision domains vs broadcast domains
  - collision: where collisions can occur
  - broadcast: where broadcasts are received

ARP (teaser):

IP → MAC on local network (detail later)

FRAME (Ethernet)

Dest MAC	Src MAC	Type	Payload	FCS
6 bytes	6 bytes	2 bytes	46-1500 bytes	4 bytes

INTERVIEW: Hub vs Switch

HUB



- operates at L1
- repeats bits to all ports
- one big collision domain
- half-duplex

SWITCH



- operates at L2
- learns MAC table (CAM)
- forwards based on MAC
- separate collision domain per port
- full-duplex (typically)

MISTAKE TO AVOID: Confusing IP (L3) with MAC (L2). IP = logical address (route globally), MAC = hardware address (local delivery).

☆ TAKEAWAY: Physical moves bits. Data link delivers on the local segment.

05 — NETWORK LAYER & IP

1) Network layer moves packets across networks — logical addressing + routing.

2) IPv4 address: 32-bit, dotted decimal
e.g. 192.168.1.10 — network part + host part.

192 . 168 . 1 . 10
network host

3) Packet header (network layer) — key fields:

Source IP	Dest IP	TTL	Protocol

← tiny view of header

4) Public vs Private IPs (RFC1918):
- 10.0.0.0/8 (10.0.0.0 – 10.255.255.255)
- 172.16.0.0/12 (172.16.0.0 – 172.31.255.255)
- 192.168.0.0/16 (192.168.0.0 – 192.168.255.255)

5) IPv6 (teaser): 128-bit address space, huge space, growing importance.

6) ICMP: ping / traceroute helpers (error & diagnostic messages).

7) Interview Q: What does TTL do?
→ TTL is hop limit. Decremented at each router. When it reaches 0, packet is dropped. Prevents infinite loops.

✗ MISTAKE: Thinking private IPs (10.x, 172.16.x, 192.168.x) are routable on the public Internet without NAT.

☆ TAKEAWAY: IP gets the packet to the right network. Routing chooses the path.

06 — SUBNETTING

① Why subnet:

- split a network into smaller pieces — smaller broadcasts, organization, security zones.

② Subnet mask / CIDR:

- 255.255.255.0 = /24 — marks network vs host bits.

③ Worked mini example:

- 192.168.1.0 /24 ⟶ 256 addresses
- Usable range: 192.168.1.1 – 192.168.1.254
- Note: 192.168.1.0 = network address (not assignable)
  192.168.1.255 = broadcast address (not assignable)

④ Formula ideas:

- If h = host bits → usable hosts ≈ $2^h - 2$
- If b = borrowed bits → number of subnets = 2^b

h = remaining host bits
b = borrowed subnet bits

⑤ Common prefix sizes (exam favorites):

CIDR	Subnet Mask	Total Addresses	Usable Hosts (approx)	Use Case (typical)
/8	255.0.0.0	16,777,216	16,777,214	Very Large Networks
/16	255.255.0.0	65,536	65,534	Enterprise / Sites
/24	255.255.255.0	256	254	Small LANs

★ Super exam favorites! ♥

⑥ Interview classic:

- Given IP + mask, find network, broadcast, usable range.

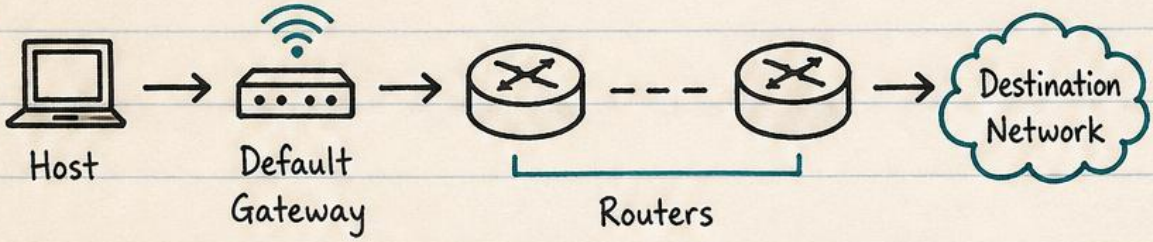
Example:
IP: 192.168.10.45 Mask: /26 (255.255.255.192)
→ Network: 192.168.10.0
→ Broadcast: 192.168.10.63
→ Usable range: 192.168.10.1 – 192.168.10.62
→ Usable hosts: 62

⚠ Common mistake:
Forgetting that network and broadcast addresses aren't assignable to hosts (IPv4 traditional).

> TAKEAWAY: Mask draws the boundary. CIDR writes it cleanly. ☆

07 — ROUTING BASICS

1. **Router** = L3 device that forwards packets between networks using a routing table.

2. 

3. **Routes**: directly connected, static, dynamic.
   - Dynamic protocols learn routes automatically.
   - Examples: OSPF (IGP), BGP (EGP).

4. **Longest prefix match** — more specific route wins.
   Example: 10.1.1.0/24 is preferred over 10.1.0.0/16 for destination 10.1.1.25.

5. **Default route** 0.0.0.0/0 — "when you don't know, go here."

6. **Interview**: Switch vs Router

SWITCH (L2)	ROUTER (L3)
• Works with MAC addresses	• Works with IP addresses
• Forwards frames within the same LAN	• Routes packets between different networks
• Does not route between networks	• Uses routing table to make decisions

7. **Mistake**: Wrong default gateway → "connected to Wi-Fi but no Internet."

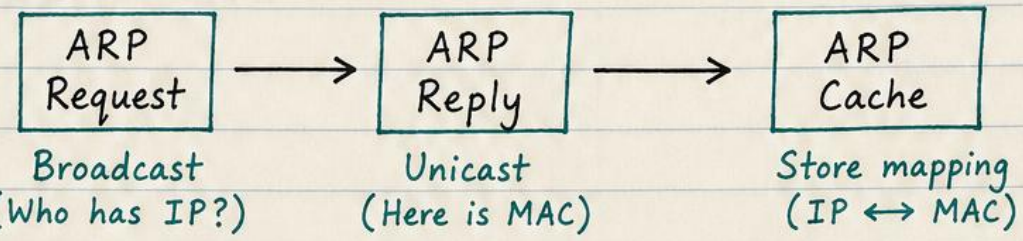
★ **TAKEAWAY**: Switches stay local. Routers cross networks.

08 — ARP & SWITCHING

① **ARP:** Address Resolution Protocol — map IP → MAC on a local network.

- Who has 192.168.1.5?
- Tell 192.168.1.10.

② Tiny sequence:



③ **Switch:** learns MAC → port from source addresses; forwards/floods intelligently.

④ **Broadcast domain** vs **Collision domain**

- Broadcast domain = where broadcasts travel.
- Collision domain = where devices can collide.

→ Switch breaks collision domains (per port).
Router breaks broadcast domains.

⑤ **Interview:** what happens when you ping a local IP first time?

1) ARP to resolve IP → MAC
2) Then ICMP Echo Request/Reply

⑥ **Mistake:** ARP spoofing awareness — attacker lies about IP–MAC mapping.

→ Leads to MITM attacks.
→ Why networks need trust controls:
DAI, DHCP Snooping, Static ARP, Port Security.

✱ **TAKEAWAY:** IP is logical. MAC is local delivery. ARP bridges them. ♡

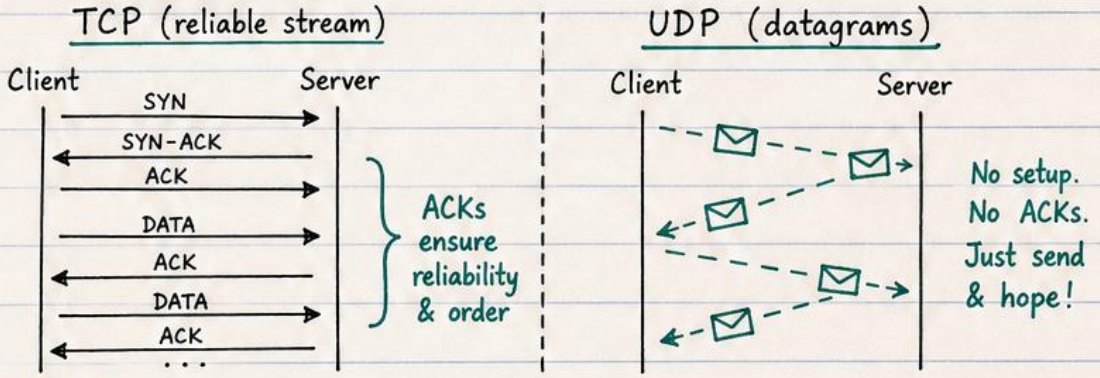
09 — TCP vs UDP

1) Transport layer: process-to-process via port numbers (0–65535).

2) Comparison table:

	TCP	UDP
Connection	Connection-oriented	Connectionless
Reliability	Reliable	Best-effort (no guarantees)
Ordering	Ordered (in-order delivery)	Not ordered
Control	Flow control + Congestion control	No flow/congestion control
Overhead	Heavier (more headers, state, handshakes)	Lightweight (smaller header, no state)
Typical uses	HTTP, SSH, email	DNS, video streaming, games, DHCP

3) How they work (tiny sketch):



4) Ports:

- Well-known ports (server side): 80 (HTTP), 443 (HTTPS), 53 (DNS), 22 (SSH)
- Ephemeral (dynamic) ports: used by clients, chosen from a high range (49152–65535) for each connection.

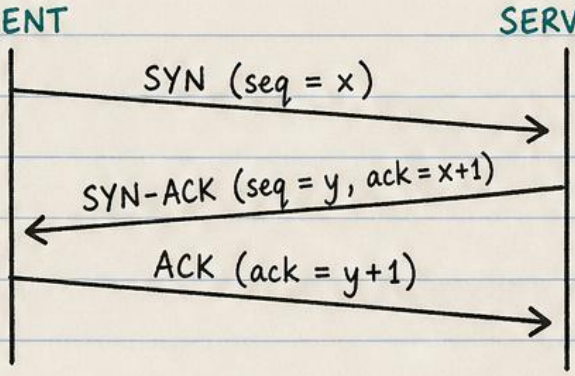
5) Interview:

- When to pick UDP: when latency is more important than perfect reliability (e.g., live video, online games, DNS). App can handle loss/retransmission.
- Mistake: "UDP is always faster and better." → WRONG!
  Reality: depends on the requirements. For reliability + order, TCP is the right choice.

✳ TAKEAWAY: TCP promises delivery, UDP promises speed of sending.

10 — TCP DEEP DIVE

1) **THREE-WAY HANDSHAKE**: SYN → SYN-ACK → ACK — establish connection (seq / ack numbers).



TEARDOWN (FOUR-WAY):

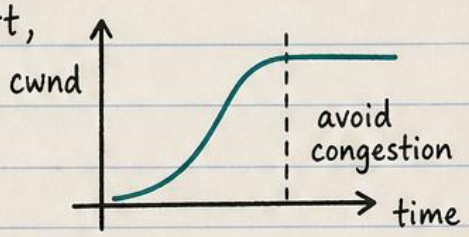
FIN →
← ACK
← FIN
→ ACK
(clean close)

2) **RELIABILITY**: sequence numbers, ACKs, retransmit on loss.

3) **FLOW CONTROL**: receiver window — don't overwhelm the receiver.



4) **CONGESTION CONTROL (IDEA)**: Slow start, avoid flooding the network. (High level)



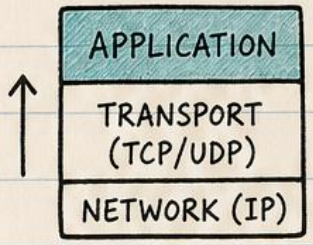
5) **INTERVIEW**: why 3-way not 2-way? — Both sides sync initial sequence numbers & readiness.

6) **MISTAKE**: assuming TCP guarantees timing — it guarantees eventual ordered delivery, not low latency.

☆ **TAKEAWAY:**
Handshake syncs. ACKs prove. Windows pace.

11 — APPLICATION LAYER

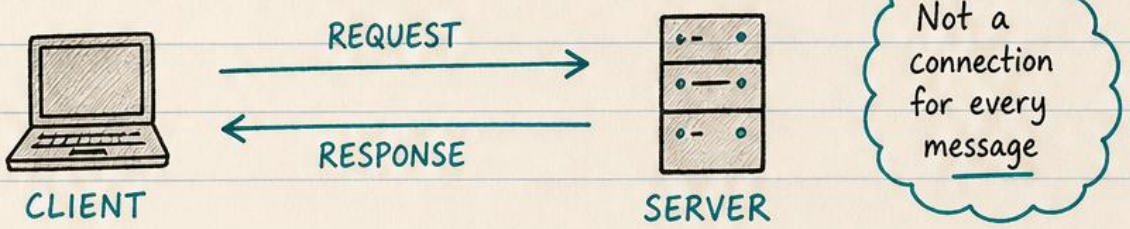
① Application protocols sit on TCP/UDP — define what messages mean.



② Protocol tour (short cards):

HTTP/HTTPS	DNS	SMTP/IMAP	SSH	FTP/SFTP	DHCP
— web	names → IPs	— mail	— secure remote shell	— files	— auto IP config

③ Client-server model (request/response):



④ Ports reminder (well-known):

80	443	53	22	25	67/68
HTTP	HTTPS	DNS	SSH	SMTP	DHCP (server/client)

⑤ Interview: "stateless HTTP"

- each request independent; server doesn't remember previous requests.
- cookies/sessions add state above HTTP.

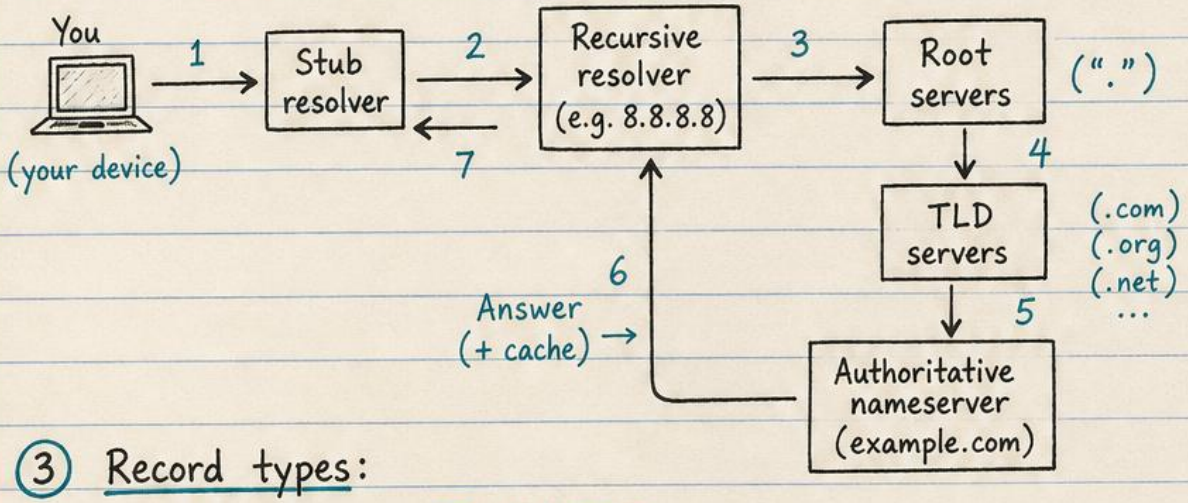
Common mistake:
Confusing application protocol with the transport under it.

TAKEAWAY:
"Apps speak protocols. Transport carries the words."

12 — DNS Deep Dive

(1) DNS = **phonebook** of the Internet — domain names → IP addresses (and more records).

(2) Resolution path sketch:

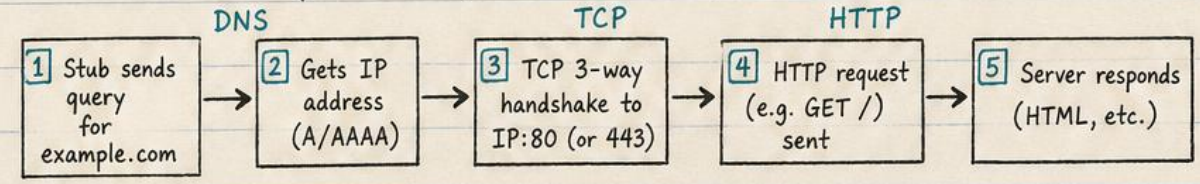


(3) Record types:

- A / AAAA → IPv4 / IPv6
- CNAME → alias to another domain
- MX → mail exchange (email routing)
- NS → nameserver (authority for zone)
- TXT → text info (SPF, verification, etc.)

(4) Usually **UDP/53**; **TCP** for large responses / zone transfers.

(5) Interview classic: trace what happens when you visit example.com (DNS then TCP then HTTP).



(6) Mistakes:

- Thinking DNS is **only A records**. (It's much more!)
- Ignoring caching / **TTL**. TTL matters for updates!

★ **TAKEAWAY:** Names for **humans**. Numbers for **routers**. DNS translates.

13 — HTTP & HTTPS

① HTTP:

- Request methods: GET / POST / PUT / DELETE
- Status codes (first digit):
  - 2xx Success
  - 3xx Redirect
  - 4xx Client error
  - 5xx Server error

Examples:
200 OK
301 Moved Permanently
404 Not Found
500 Internal Server Error

② Tiny request sketch:

METHOD PATH VERSION	← request line
HEADERS (key: value)	← headers
(optional) BODY	← optional body (e.g. JSON)

③ HTTPS = HTTP over TLS

- Provides: encryption, integrity, server authenticity (certificates).

④ TLS handshake idea (high level):

1. ClientHello (versions, ciphers, random)
2. ServerHello + Certificate (server identity)
3. Key exchange / both derive session keys
4. Handshake finished → Encrypted HTTP from now on

Goal: agree on keys & verify the server, then lock the channel.

⑤ HTTP/1.1 vs HTTP/2 (brief):

- HTTP/1.1: one request at a time per connection (or a few with multiple connections).
- HTTP/2: multiplexing → many requests/responses over the same TCP connection, reduces head-of-line blocking.

⑥ Interview note:

- HTTPS doesn't hide the URL path from your ISP the same way as a full VPN.
- TLS protects content in transit; SNI (Server Name Indication) and DNS lookups can reveal what site you're connecting to.

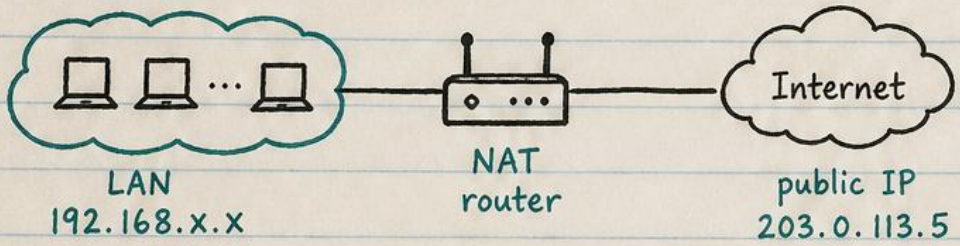
⑦ Common mistake:

- "HTTP1 means the website is safe / trustworthy" — Not exactly. HTTPS only means the channel is encrypted & the server's identity is verified (if cert is valid). It does NOT guarantee the website or its content is safe / trustworthy.

TAKEAWAY: HTTP is the web's language. TLS locks the envelope.

14 — NAT, FIREWALL & PROXY

① **NAT**: many private hosts share public IPs — translate addresses/ports at the edge. Enables private addressing.

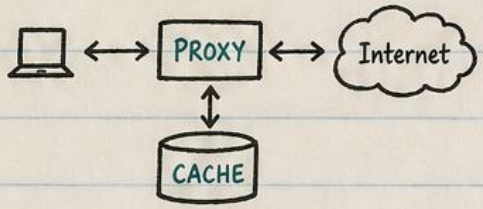


② **FIREWALL**: allow/deny traffic by rules (ports, IPs, state).

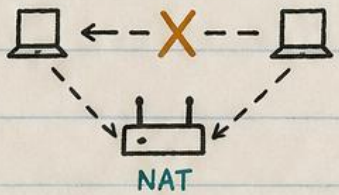
Packet filter vs stateful.

ACTION	SRC IP	DST IP	PORT	STATE
ALLOW	10.0.0.0/8	ANY	80,443	ANY
DENY	ANY	10.0.0.0/8	23	NEW
ALLOW	ANY	ANY	ESTABLISHED	EST.

③ **PROXY**: intermediary for requests — caching, filtering, anonymity patterns.



④ **INTERVIEW**: why NAT complicates peer-to-peer — inbound connections need mapping / port forward / STUN.



⑤ **MISTAKE**: equating NAT with security — helpful obscurity, not a full firewall.

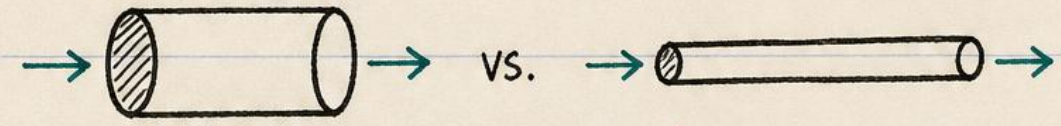
NAT ≠ SECURITY

TAKEAWAY: NAT remaps. Firewalls decide. Proxies stand in the middle.

15 — Performance Basics

1) **Bandwidth** = capacity (bits/sec).
   Latency = delay (time).
   Throughput = actual delivery rate.

2) Tiny sketch:



fat pipe (high bandwidth) vs. long pipe (high latency)

Both matter! Data in flight ≈ Bandwidth × Latency (bandwidth-delay product)

3) **Congestion**: too much traffic → queues → loss → TCP backs off.

4) **Jitter**: variation in delay — hurts real-time audio/video.

5) **QoS teaser**: prioritize some traffic (VoIP) over bulk.

6) **Interview**: ping measures RTT; traceroute shows path hops.

7) **Mistake**: upgrading bandwidth alone when the problem is latency or packet loss.

TAKEAWAY: Capacity, delay, loss — measure the right one.

16 — SECURITY ESSENTIALS

1) THREATS:

- eavesdropping — someone listens to your traffic.
- spoofing — attacker pretends to be someone else.
- MITM (Man-in-the-Middle) — attacker sits between you and the real party.
- DDoS — flood target with traffic; make it unavailable.
- malware on the wire path — sniff, modify, steal.

2) DEFENSES:

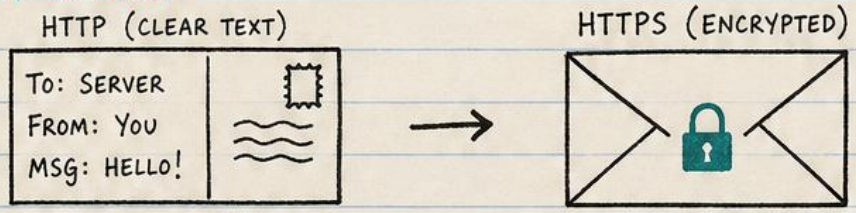
- TLS/HTTPS — encrypt data in transit.
- Firewalls — control what comes in and out.
- VPN tunnels — encrypt across untrusted networks.
- Authentication — verify who is who.
- Least privilege on network access — only what's needed.

3) CIA TRIAD:

- Confidentiality — keep data secret.
- Integrity — keep data accurate and unmodified.
- Availability — keep services up and reachable.

Networks touch all three.

4) TINY SKETCH:



Like a postcard.
Anyone can read it.

Like a sealed envelope.
Only the server can open it.

5) INTERVIEW:

VPN vs HTTPS —

- VPN tunnels many apps across the network (your whole traffic).
- HTTPS protects one session to a website (e.g., example.com).

6) MISTAKE TO AVOID:

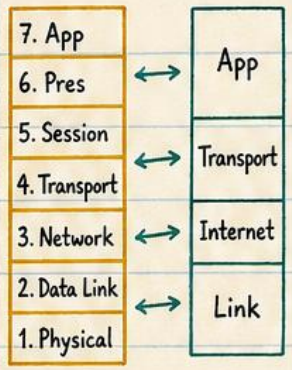
- Open Wi-Fi + HTTP = readable traffic. Anyone nearby can see it.
- Weak/default passwords on routers = easy entry for attackers.

TAKEAWAY: Encrypt in transit. Filter at edges. Assume hostile networks.

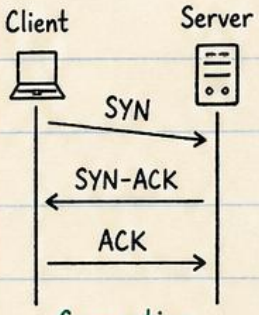
17 — CLASSIC INTERVIEW Qs

PATTERN BANK

1 Explain OSI vs TCP/IP



2 TCP 3-way handshake



Connection Established!

3 TCP vs UDP when to use

TCP

- Reliable
- Ordered
- Use: Web, Email, File Transfer

UDP

- Fast, No guarantee
- Use: Streaming, Gaming, DNS

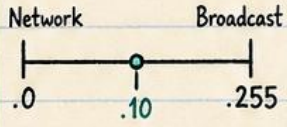
4 What happens when you type a URL

(DNS → TCP → TLS → HTTP)

1. DNS Lookup
2. TCP Connection
3. TLS Handshake
4. HTTP Request/ Response

5 Subnet: find network / broadcast

IP: 192.168.1.10/24



Network: 192.168.1.0

Broadcast: 192.168.1.255

Usable: 192.168.1.1 – 192.168.1.254

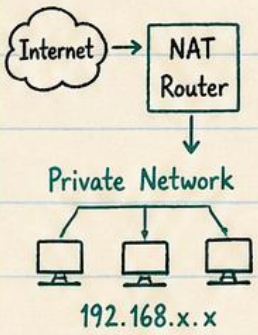
6 Switch vs Router vs Hub

Hub: Repeats to all ports. No intelligence.

Switch: L2 device. Forwards using MAC table.

Router: L3 device. Forwards using IP addresses. Connects different networks.

7 NAT purpose



Allows many private devices to share one public IP.

8 ARP role

Maps IP → MAC on the local network.



Used before sending a frame to the next hop.

APPROACH: clarify layer → draw path → name protocols → edge cases (cache, failure).

TIP: narrate the packet's journey layer by layer.

MISTAKE: dumping acronyms without causality.

18 – CHEAT SHEET

① OSI 7 LAYERS – one-liners

7	Application	User services, network apps (HTTP, DNS, SMTP)
6	Presentation	Data format, encryption, compression
5	Session	Establish, manage, terminate sessions
4	Transport	End-to-end delivery, segmentation, ports
3	Network	Logical addressing, routing, path selection
2	Data Link	Framing, MAC addressing, error detection
1	Physical	Bits on wire, signaling, cables, connectors

② TCP/IP 4-LAYER MAP

4	Application (HTTP, DNS, SMTP, etc.)	= OSI 7,6,5
3	Transport (TCP, UDP)	= OSI 4
2	Internet (IP, ICMP)	= OSI 3
1	Network Access (Ethernet, Wi-Fi, etc.)	= OSI 2,1

③ COMMON PORTS

20	FTP (data)	53	DNS
21	FTP (control)	80	HTTP
22	SSH	443	HTTPS
25	SMTP	67/68	DHCP (server/client)

④ PRIVATE IPv4 RANGES

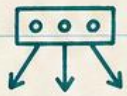
- 10.0.0.0/8 (10.0.0.0 – 10.255.255.255)
- 172.16.0.0/12 (172.16.0.0 – 172.31.255.255)
- 192.168.0.0/16 (192.168.0.0 – 192.168.255.255)

Not routable on the public Internet.

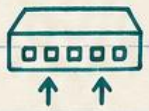
⑤ TCP vs UDP

TCP	UDP
• Connection-oriented	• Connectionless
• Reliable delivery	• Best-effort delivery
• Ordered delivery	• No ordering guarantee
• Ack + Retransmission	• No ack / retransmission
• Flow control	• No flow control
• Higher overhead	• Lower overhead
• Use: Web, Email, SSH	• Use: DNS, VoIP, Video

⑥ NETWORK DEVICES



HUB (L1)
Repeats bits to all ports.
No intelligence. Legacy.



SWITCH (L2)
Uses MAC addresses.
Forwards only where needed.



ROUTER (L3)
Uses IP addresses.
Routes between networks.

⑦ CIDR QUICK REFERENCE

/24 = 255.255.255.0 (256 addrs, 254 usable)
/16 = 255.255.0.0 (65,536 addrs)
/8 = 255.0.0.0 (16,777,216 addrs)

Usable hosts = 2^(32 - prefix) – 2
(subtract network & broadcast)

⑧ HTTP STATUS CODES

2xx	Success	(200 OK, 204 No Content)
3xx	Redirection	(301 Moved, 302 Found)
4xx	Client Error	(404 Not Found, 403 Forbidden)
5xx	Server Error	(500 Internal Server Error)

⑨ MENTAL CHECKLIST



WHICH LAYER?
What's the problem?
Map it to the OSI / TCP-IP layer.
Start there.



WHICH ADDRESS?
- L2 (Data Link): MAC
- L3 (Network): IP
- L4 (Transport): Port

Pick the right one.



RELIABLE NEEDED?
Need guaranteed delivery and order? → Use TCP
Speed > Reliability?
→ Use UDP

19 — ROADMAP & RECAP

1) PATH:

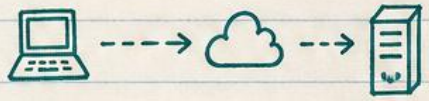
① Models (OSI/TCP-IP) → ② L1/L2 → ③ IP & subnet → ④ Routing/ARP → ⑤ TCP/UDP → ⑥ DNS/HTTP → ⑦ NAT/security → ⑧ drill interviews

2) BIG TAKEAWAY:

> Think in layers.
> Address at the right layer.
> Pick TCP when truth matters,
> UDP when time matters.

3) PRACTICE LOOP:

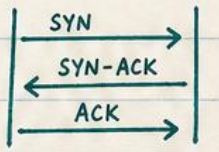
- Draw URL journey from scratch.



- Subnet 5 problems.



- Explain handshake aloud.



4) NEXT TOPICS (AFTER FUNDAMENTALS STICK):

Wireshark labs → BGP deep dive → load balancers → CDN

(one step at a time!)

5) EXAM TIP:

☆ Diagrams score — always sketch the path.

6) CLOSING:

♡ @codedsoul_05 — connect with intent.